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TOTAL						100

Objective Assessment Questions	Level	Mark Weightage (%)	Course Outcome	Mark (%)
		25		
		50		
		25		
			TOTAL	
			Course Learning Outcome	

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Exp./Prog.No:	3
Name of the Exp./ Prog.	To perform analysis on digital signal using Spectrum Analyzer and digital oscilloscope and verify effects on digital characteristics.
Date of the Exp.	
CO	CO 1
CO/PO/PSO Mapped to the Exp.	

Objectives	Instruments Required	Procedure	Observations	Inferences	Applications	Viva Questions	Worksheet
To perform analysis on digital signal using Spectrum Analyzer and verify effects on digital characteristics	Digital Oscilloscope, DSO, DSO Interconnecting cables, Spectrum Analyzer, Function Generator, Power Supply, Digital Trainer Kit, Connecting wires	Refer to the manual of the equipment carefully, Switch on the equipment, Set the controls, Connect the input signal, Observe the output waveform	Observe the waveform, Note down the readings	Explain the observations	Discuss the applications	Answer the viva questions	21
To perform analysis on digital signal using Spectrum Analyzer and verify effects on digital characteristics	Digital Oscilloscope, DSO, DSO Interconnecting cables, Spectrum Analyzer, Function Generator, Power Supply, Digital Trainer Kit, Connecting wires	Refer to the manual of the equipment carefully, Switch on the equipment, Set the controls, Connect the input signal, Observe the output waveform	Observe the waveform, Note down the readings	Explain the observations	Discuss the applications	Answer the viva questions	22
To perform analysis on digital signal using Spectrum Analyzer and verify effects on digital characteristics	Digital Oscilloscope, DSO, DSO Interconnecting cables, Spectrum Analyzer, Function Generator, Power Supply, Digital Trainer Kit, Connecting wires	Refer to the manual of the equipment carefully, Switch on the equipment, Set the controls, Connect the input signal, Observe the output waveform	Observe the waveform, Note down the readings	Explain the observations	Discuss the applications	Answer the viva questions	23

Objective / Assessment Criteria	Level	Max Weightage (%)	Mid Obtained	Viva Obt.
Explores applications using, APIs, external software tools		25		
Analyzes design problems by applying fundamentals		15		
Compares traditional solutions with functional results		5		
Interprets results and documents findings		5		
		Total		
Course Faculty Signature				

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Subjects Associated Laboratory	Level	Max Weightage (%)	Work Planned	Work Done
Communication and Presentation Strategies		5		
Project preliminary and Validation		5		
Technological Innovation Entrepreneurship		5		
Knowledge Management and Patents		5		
UNPAID Internship Industrial Exposure		5		
		Total		
		Course Faculty Signature		

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[illegible] Outcomes	Level	Mapping with CO	Mapping with PSO	Mapping with PO
Demonstrate the principles of [illegible]		1		
Analyze [illegible]		2		
Apply [illegible]		3		
Demonstrate [illegible]		4		
Apply [illegible]		5		
Total				
Course Faculty Signature				

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	Example: MATLAB				

Objective / Assessment Criterion	Level	Max Weightage (%)	Mark Obtained	Mark (10)
			Total	
			Course Faculty Signature (If Any)	

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[illegible]	11
[illegible]	[illegible]
[illegible]	
[illegible]	00.00.00
[illegible]	

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
CO3 compression and digital representation	[illegible]	[illegible]	[illegible]	[illegible]	11
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	11
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	11

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Objective / Outcome / Criteria	Level	Weightage (%)	Weighted Value	Attainment
CO1 Knowledge and clinical representation		3		
Medical and physiological models		3		
Factors influencing property modification		3		
Biomedical and material analysis		3		
PATLAB provides clinical and engineering interpretation		3		
		Total		
		Course Faculty (Sign with date)		

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[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Objective / Learning outcome	Level	Bloom's Taxonomy Level (BL)	Exam Weightage (%)	Mark (Out of 100)
CLO1		L2		
CLO2		L3		
CLO3		L3		
CLO4		L4		
CLO5		L3		
Total				

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Experiment No.	9
Name of the Experiment	To determine the Biological Oxygen Demand (BOD) of Liquid Waste Samples and Analyze its significance in Environmental Engineering Using Standard Procedures
Date of the Experiment	
CO	CO5
Name of the Team Members of the Experiment	

Performance Parameter	Excellent - 3	Good - 2	Satisfactory - 1	Needs Improvement - 0	Marks Obtained
Understanding the importance of BOD in wastewater analysis	Demonstrates a clear understanding of the importance of BOD in wastewater analysis and its role in environmental engineering	Demonstrates a good understanding of the importance of BOD in wastewater analysis, but may have minor gaps in knowledge	Demonstrates a basic understanding of the importance of BOD in wastewater analysis, but has significant gaps in knowledge	Does not demonstrate an understanding of the importance of BOD in wastewater analysis	
Accuracy of BOD measurements and calculations	Accurately measures BOD and performs calculations with no errors	Measures BOD and performs calculations with minor errors	Attempts to measure BOD and perform calculations, but with significant errors	Does not attempt to measure BOD or perform calculations	
BOD data interpretation and analysis	Correctly interprets BOD data and draws accurate conclusions about the wastewater sample	Interprets BOD data with minor errors in conclusions drawn	Attempts to interpret BOD data, but with significant errors in conclusions drawn	Does not attempt to interpret BOD data or draw conclusions	
Safety and environmental considerations	Demonstrates a thorough understanding of safety and environmental considerations when handling wastewater samples	Demonstrates a good understanding of safety and environmental considerations when handling wastewater samples	Demonstrates a basic understanding of safety and environmental considerations when handling wastewater samples	Does not demonstrate an understanding of safety and environmental considerations when handling wastewater samples	

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[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

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Objective/ Assessment Criterion	Level	Max. Weightage (%)	Marks (Out of 100)	Marks (/10)
		20		
		10		

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S. No.	Part	Marks obtained/Expressed	Signature	Remarks	Signature of Invigilator
1					